

Experiment 6: Depth First Search (DFS)

Aim

To implement Depth First Search (DFS) using Python.

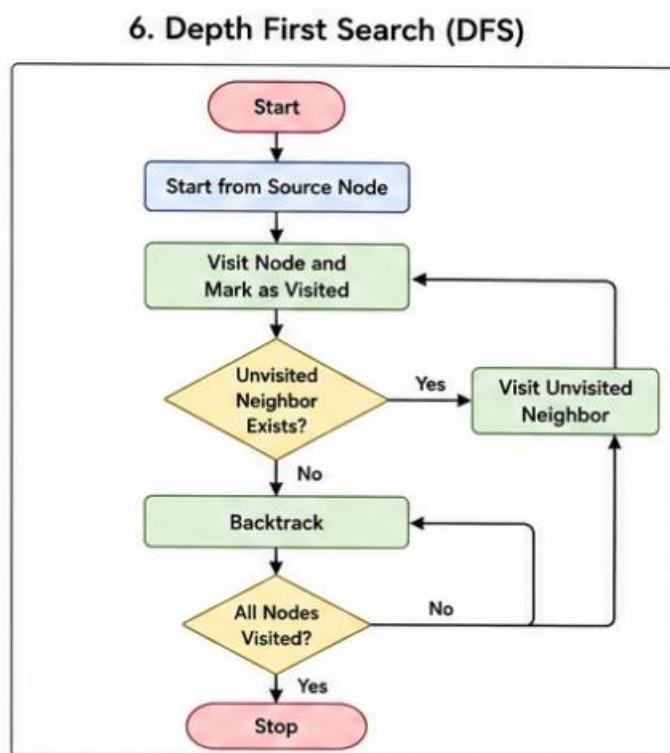
Objective

To traverse all nodes of a graph using DFS.

Algorithm

1. Create a graph.
2. Start from the source node.
3. Visit the node.
4. Recursively visit all unvisited neighbours.
5. Repeat until all nodes are visited.

Flowchart



Code

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],
```

```

'C':['F'],
'D':[],
'E':[],
'F':[]
}

```

```
visited = set()
```

```
def dfs(node):
```

```
    if node not in visited:
```

```
        print(node, end=" ")
```

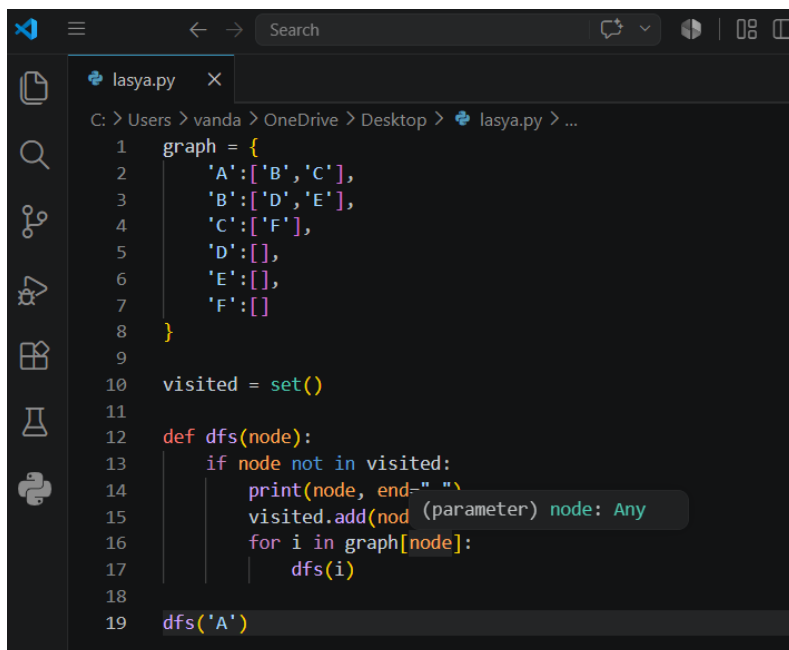
```
        visited.add(node)
```

```
        for i in graph[node]:
```

```
            dfs(i)
```

```
dfs('A')
```

Output:Result



```

1  graph = {
2      'A':['B','C'],
3      'B':['D','E'],
4      'C':['F'],
5      'D':[],
6      'E':[],
7      'F':[]
8  }
9
10 visited = set()
11
12 def dfs(node):
13     if node not in visited:
14         print(node, end=" ")
15         visited.add(node)
16         for i in graph[node]:
17             dfs(i)
18
19 dfs('A')

```

```

PS C:\Users\vanda\onedrive\desktop> python lasya.py
A B D E C F
PS C:\Users\vanda\onedrive\desktop>

```

The graph was traversed successfully using DFS.

Conclusion

DFS visits nodes by exploring each branch completely before backtracking.